Summary in Graph

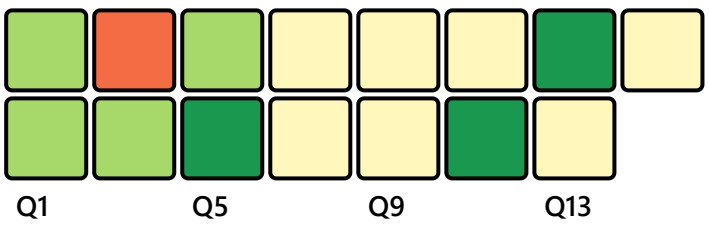
# Exam Summary (GO Classes CS Test Series 2026 | Databases | Topic Wise Test 5)

| | |
|---|---|
| Qs. Attempted: | 12 (5 + 7) |
| Correct Attempts: | 7 (4 + 3) |
| Incorrect Attempts: | 5 (1 + 4) |
| Correct Marks: | 10 (4 + 6) |
| Penalty Marks: | 0.33 (0.33 + 0) |
| Resultant Marks: | 9.67 (3.67 + 6) |

| | |
|---|---|
| Total Questions: | 15 (5 + 10) |
| Total Marks: | 25 (5 + 20) |
| Exam Duration: | 45 Minutes |
| Time Taken: | 40 Minutes |

## Individual Question Statistics

Select Metric: Marks Obtained




* Average Stats based on first-attempt stats of Top 10% (124) out of 1242 users

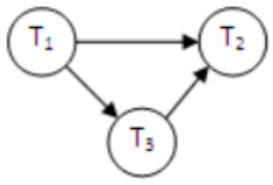
EXAM RESPONSE EXAM STATS FEEDBACK

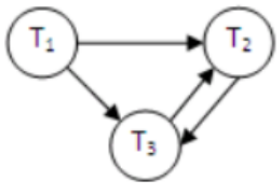
# Technical

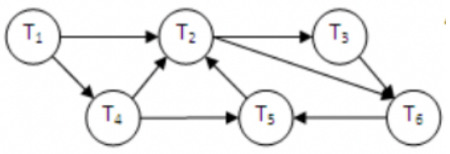
Q #1 Multiple Select Type Award: 1 Penalty: 0 Databases

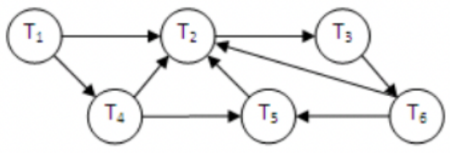
A simple way to detect a state of deadlock is to construct a wait-for graph. One node is created in the graph for each transaction that is currently executing in the schedule. Whenever a transaction $\mathrm{T}_i$ is waiting to lock an item $\mathrm{X}$ that is currently locked by a transaction $\mathrm{T}_j$, it creates a directed edge $(\mathrm{T}_i \# \mathrm{T}_j)$. When $\mathrm{T}_j$ releases the lock(s) on the items that $\mathrm{T}_i$ was waiting for, the directed edge is dropped from the waiting-for graph.

Given the graph below, which is/are deadlock free?

A. 


B. 


C. 


D. 


Your Answer: A | Correct Answer: A | Correct | Time taken: 01min 17sec | Discuss

**Q #2** Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Which of the following statements is/are true?

A. Serializability is the property that a (possibly interleaved) execution of a group of transactions has the same effect on the database and produces the same output as some serial execution of those transactions.

B. The following schedule is serializable:
$\mathrm{r1[X] \rightarrow w1[X] \rightarrow r2[Y] \rightarrow w2[Y] \rightarrow r2[X] \rightarrow w2[X] \rightarrow r1[Z] \rightarrow w1[Z] \rightarrow c1 \rightarrow c2}$
Here in $\mathrm{T\#[A]}$, $\mathrm{T}$ represents operation(read/write), $\#$ represents transaction number, $\mathrm{A}$ represents the item on which operation is being performed.

C. Strict two-phase locking $(\mathrm{2PL})$ ensures that transactions never deadlock.

D. Strict two-phase locking $(\mathrm{2PL})$ ensures conflict serializability.

Your Answer: A;B;D | Correct Answer: A;B;D | Correct | Time taken: 02min 01sec | Discuss

**Q #3** Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Assume by "Serializability", we mean "Conflict Serializability".
For the given schedule, which of the following is TRUE?

| $\mathrm{T_1}$ | $\mathrm{T_2}$ | $\mathrm{T_3}$ | $\mathrm{T_4}$ |
|---|---|---|---|
| | | | Read(A) |
| | Read(A) | | |
| | | Read(A) | |
| Write(B) | | | |
| | Write(A) | | |
| | | Read(B) | |
| | Write(B) | | |

A. The above schedule is not serializable

B. The above schedule is equivalent to serial schedule $\mathrm{T_3, T_4, T_1, T_2}$

C. The above schedule is equivalent to serial schedule $\mathrm{T_1, T_4, T_3, T_2}$

D. The above schedule is equivalent to serial schedule $\mathrm{T_2, T_3, T_1, T_4}$

Your Answer: B | Correct Answer: C | Incorrect | Time taken: 01min 39sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

To check for conflict serializability of any schedule, we draw its Precedence graph. In the precedence graph we have a node for each transaction $\mathrm{T}_i$ and An edge from $\mathrm{T}_i$ to $\mathrm{T}_j$ whenever an action in $\mathrm{T}_i$ conflicts with, and comes before an action in $\mathrm{T}_j$.

For a conflict serializable schedule $\mathrm{S}$, which of the following is guaranteed to give a serial schedule that is conflict equivalent to $\mathrm{S}$, in the precedence Graph?

A. Depth-first order
B. Topological order
C. Breadth-first order
D. None

Your Answer: B | Correct Answer: B | Correct | Time taken: 00min 11sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Consider the following schedules:

| S1 | | S2 | | S3 | | S4 | |
|---|---|---|---|---|---|---|---|
| T1 | T2 | T1 | T2 | T1 | T2 | T1 | T2 |
| R(A) | | | R(B) | R(A) | | | R(B) |
| W(A) | | | W(A) | W(A) | | R(A) | |
| | R(B) | R(A) | | | R(B) | | W(A) |
| R(B) | | W(A) | | | W(A) | W(A) | |
| W(B) | | R(B) | | R(B) | | R(B) | |
| | W(A) | W(B) | | W(B) | | W(B) | |
| | R(C) | | R(C) | R(C) | | | R(C) |
| R(C) | | R(C) | | | R(C) | R(C) | |

Which of the above schedule(s) is/are conflict serializable?

A. $\mathrm{S1}$ and $\mathrm{S2}$ only
B. $\mathrm{S2}$ only
C. $\mathrm{S1}$ only
D. None of these

Your Answer: B | Correct Answer: B | Correct | Time taken: 01min 12sec | Discuss

Q #6 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Suppose that transaction $\mathrm{T1}$ aborts after the following sequence of read write operations in the given schedule below :

$$\mathrm{Write1(X);\ Write2(X);\ Write1(Y);\ Read3(Y);\ Read\,4(X);\ Read\,4(Z)}$$

Which of the other transactions would need to be rolled back?

A. $\mathrm{T2}$
B. $\mathrm{T3}$
C. $\mathrm{T4}$
D. None

Your Answer: B | Correct Answer: B | Correct | Time taken: 01min 25sec | Discuss

Q #7 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider the following concurrent execution of three transactions $\mathrm{T}_1, \mathrm{T}_2, \mathrm{T}_3$ :

| $\mathrm{T}_1$ | $\mathrm{T}_2$ | $\mathrm{T}_3$ |
|---|---|---|
| $w(x)$ | | |
| | $r(y)$ | |
| | | $w(z)$ |
| $w(y)$ | | |
| | $r(z)$ | |
| | | $w(z)$ |

- $w(\mathrm{P})$ means Write operation on "P"
- $r(\mathrm{P})$ means Read operation on "P"

Which of the following is/are False about the given schedule?

A. It is conflict serializable.
B. It is view serializable.
C. It can be produced by a Two Phase Lock $(2\mathrm{PL})$ scheduler.
D. It can be produced by a strict $2\mathrm{PL}$ scheduler.

Your Answer: A;B | Correct Answer: A;B;C;D | Incorrect | Time taken: 09min 48sec | Discuss

Q #8 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

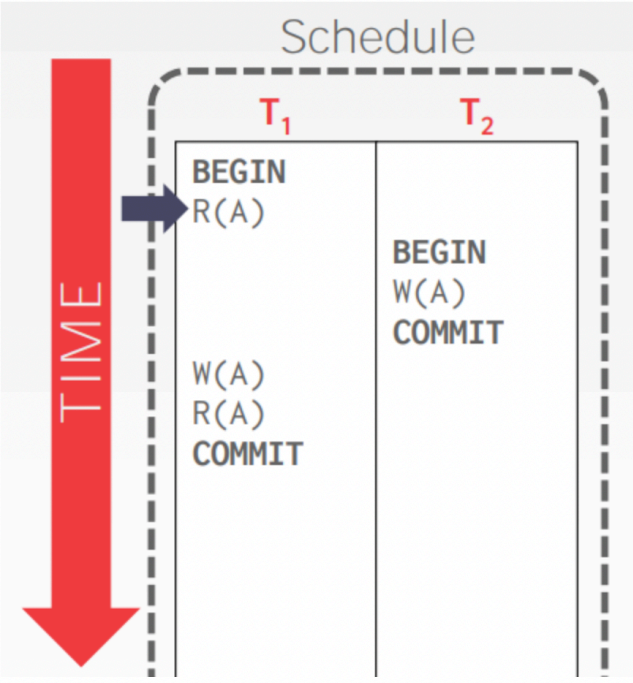
Consider a database schedule of two transactions $\mathrm{T1}, \mathrm{T2}$. $\mathrm{T1}$ first reads $\mathrm{A}$ and $\mathrm{B}$ and then writes $\mathrm{A}$ and $\mathrm{B}$. $\mathrm{T2}$ reads and writes $\mathrm{A}$ then reads and writes $\mathrm{B}$. Consider the following statements. Which of the following statements are correct?

A. Every concurrent/interleaved execution of these transactions will lead to a conflict serializable schedule.
B. Every concurrent/interleaved execution of these transactions will lead to a view serializable schedule.
C. If a concurrent/interleaved execution of these transactions lead to a non-conflict serializable schedule then that schedule is also non-view serializable.
D. If a concurrent/interleaved execution of these transactions lead to a non-view serializable schedule then that schedule is also non-conflict serializable.

Your Answer: B;D | Correct Answer: C;D | Incorrect | Time taken: 05min 17sec | Discuss

Q #9 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider the following schedule $\mathrm{S}$ :




The time-stamps of $\mathrm{T1}, \mathrm{T2}$ are $1, 2$ respectively then which of the following is/are Incorrect?

A. $\mathrm{S}$ is possible under basic Timestamp Ordering (T/O) protocol.
B. $\mathrm{S}$ is possible under THOMAS WRITE RULE Timestamp Ordering (T/O) protocol.
C. $\mathrm{S}$ is conflict serializable.
D. $\mathrm{S}$ is view serializable.

Your Answer: A;B | Correct Answer: A;B;C;D | Incorrect | Time taken: 01min 24sec | Discuss

**Q #10** Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider the following schedule $\mathrm{S}$ :

$$\mathrm{start1;\ start2;\ r1(A);\ r2(B);\ w2(A);\ w1(B);}$$

"$\mathrm{start}\ \#$" means beginning of transaction $\#$, then which of the following is/are Incorrect?

A. $\mathrm{S}$ is possible under basic Timestamp Ordering (T/O) protocol.
B. $\mathrm{S}$ is possible under THOMAS WRITE RULE Timestamp Ordering (T/O) protocol.
C. $\mathrm{S}$ is conflict serializable.
D. $\mathrm{S}$ is view serializable.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 29sec | Discuss

**Q #11** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following schedule:

$$\mathrm{S} = (2, \mathrm{W}, \mathrm{x}),(1, \mathrm{W}, \mathrm{z}),(1, \mathrm{W}, \mathrm{x}), \mathrm{C1},(2, \mathrm{W}, \mathrm{y}),(2, \mathrm{R}, \mathrm{y})(3, \mathrm{W}, \mathrm{z}),(3, \mathrm{W}, \mathrm{m}), \mathrm{C3},(2, \mathrm{R}, \mathrm{m}), \mathrm{C2}.$$

Where $\mathrm{Ci}$ means commit of $\mathrm{i}^{\mathrm{th}}$ transaction. Triple $(\mathrm{i}, \mathrm{W} / \mathrm{R}, \mathrm{x})$ means Transaction $\mathrm{i}$ writes/reads item $\mathrm{x}$. Which of the following statements is/are true for $\mathrm{S}$?

A. $\mathrm{S}$ is conflict serializable.
B. $\mathrm{S}$ is view serializable but not conflict serializable.
C. $\mathrm{S}$ is not view serializable.
D. None

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 05min 04sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following schedule :

| $T_1$ | $T_2$ | $T_3$ | $T_4$ |
|---|---|---|---|
| $W(a)$ | | | |
| | $R(a)$ | | |
| | | $W(a)$ | |
| $W(a)$ | | | |
| | | | $W(a)$ |

Which of the following is/are true for this schedule?

A. It is conflict serializable but not view serializable.
B. It is view serializable but not conflict serializable.
C. It is neither conflict nor view serializable.
D. It is both conflict and view serializable.

Your Answer: C | Correct Answer: C | Correct | Time taken: 01min 11sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following schedule $S$ :

$$\begin{array}{lllllll} T_1: & & & r_1(A) & & w_1(B) & \\ T_2: & r_2(B) & w_2(A) & & & & w_2(B) \\ T_3: & & & & r_3(A) & & & w_3(B) \end{array}$$

Which of the following is/are true for this schedule?

A. It is conflict serializable but not view serializable.
B. It is view serializable but not conflict serializable.
C. It is neither conflict nor view serializable.
D. It is both conflict and view serializable.

Your Answer: B | Correct Answer: B | Correct | Time taken: 02min 33sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Let transactions $T1$, $T2$ and $T3$ be defined to perform the following operations :

- $T1$ : Add one to $A$
- $T2$ : Double $A$
- $T3$ : Display $A$ on the screen and then set it to $1$ ie. $A = 1$.

(where $A$ is some item in the database)
Suppose transactions $T1$, $T2$ and $T3$ are allowed to execute concurrently. If $A$ has initial value zero, report the possible values of final value of $A$ and the displayed value of $A$.

A. $1, 2$
B. $2, 1$
C. $3, 0$

D. $4, 0$

Your Answer: B;C;D | Correct Answer: A;B;C;D | Incorrect | Time taken: 05min 00sec | Discuss

**Q #15** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

In the following, when a transaction $\mathrm{T}$ performs an operation $x = x + a$, then $\mathrm{T}$ reads $x$ into some register, adds a to the register content then stores register content into $x$.

A transaction schedule is serializable if its effect is equivalent to that of some serial schedule. Consider a bookkeeping operation consisting of two transactions $-\mathrm{T}_1$ and $\mathrm{T}_2-$ that are required to keep the sum $\mathrm{A}+\mathrm{B}+\mathrm{C}$ unchanged. Which of the following pairs of transactions will always result in a serializable schedule?

I.

| $T_1$ | $T_2$ |
|---|---|
| Lock A; | Lock B; |
| A = A - 10; | B = B - 20; |
| Unlock A; | Unlock B; |
| B = B + 10; | C = C + 20; |

II.

| $T_1$ | $T_2$ |
|---|---|
| A = A - 10; | Lock B; |
| Lock B; | B = B - 20; |
| B = B + 10; | Unlock B; |
| Unlock B; | C = C + 20; |

III.

| $T_1$ | $T_2$ |
|---|---|
| Lock A; | Lock A; |
| A = A - 10; | B = B - 20; |
| Unlock A; | Unlock A; |
| B = B + 10; | C = C + 20; |

A. I only
B. II only
C. III only
D. I and II

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 01min 27sec | Discuss

Copyright & Stuff